

SHIVANSH THAKUR

APPLIED AI ENGINEER · COMPUTER VISION · OCR & DOCUMENT INTELLIGENCE

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SUMMARY

Applied AI Engineer focused on computer vision, OCR and local AI systems. Hands-on with YOLOv8, PaddleOCR, Whisper, DistilBERT and real-time inference pipelines in Python and PyTorch. Built production-focused systems across ALPR, voice AI assistants and document-intelligence workflows. MCA in AI & ML, Chandigarh University.

KEY HIGHLIGHTS

- Built a real-time ALPR pipeline combining YOLOv8 detection, SORT tracking and PaddleOCR for traffic video.
- Developed a low-latency local voice AI assistant (Whisper + Ollama + Piper TTS over LiveKit) with sub-1s interruption handling.
- Fine-tuned a DistilBERT classifier for multilingual (English / Hindi / Hinglish) customer message routing across 5 business intents.
- Shipped a production OCR pipeline extracting 7 fields per consignment note in ~10 seconds on CPU.
- Led RFM segmentation on 4,300+ customers, surfacing ~₹8.3M in revenue clusters during data-science internship.

TECHNICAL SKILLS

AI / ML Python, PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, DistilBERT, CNNs, AutoML (PyCaret), Class-Weighted Training

Computer Vision & OCR YOLOv8, SORT Tracker, OpenCV, PaddleOCR, Tesseract, MediaPipe FaceMesh, solvePnP head-pose, Image Preprocessing

Voice AI & LLMs LiveKit (WebRTC), Whisper (STT), Piper TTS, Silero VAD, Ollama, Local LLMs / GGUF, Real-time Streaming

Data & BI SQL, Pandas, NumPy, EDA, Feature Engineering, RFM / CLTV, Statistical Analysis, Power BI, DAX, Tableau

Tooling Streamlit, Git, GitHub, CUDA / GPU Inference, Linux, JIRA, Agile, YAML-driven config

WORK EXPERIENCE

AI Engineer — Computer Vision & OCR

Nov 2025 – Present

Harijsoftech

On-site, India

- **Voice AI** • Built a low-latency local voice AI assistant using Whisper (STT), Silero VAD, Ollama (LLM) and Piper TTS over LiveKit WebRTC, with sub-1s interruption handling and two reasoning modes (chat streaming + JSON tool-calls); CUDA-accelerated, no cloud APIs. [GitHub ↗](#)
- **ALPR** • Developed a real-time ALPR pipeline combining YOLOv8 detection, SORT multi-vehicle tracking and PaddleOCR; added a per-vehicle character-voting layer to stabilise OCR reads across frames; validated on UK and Indian traffic footage. [GitHub ↗](#)
- **DistilBERT Router** • Fine-tuned a DistilBERT WhatsApp intent router for multilingual (English / Hindi / Hinglish + emojis) customer messages across 5 business intents; lifted accuracy from a ~50% baseline to production-ready levels by redesigning the label taxonomy (8 → 5 cleanly-separable classes) and applying class-weighted training. [GitHub ↗](#)
- **Consignment OCR** • Shipped a production OCR pipeline for consignment / dispatch notes using PaddleOCR, OpenCV and YAML-driven configuration — extracts 7/7 fields in ~10 seconds on CPU; bounding boxes auto-scale across phone, flatbed and document-camera scans; batch mode, logging and test suite included. [GitHub ↗](#)
- **RC Extractor** • Built a rule-based Indian Vehicle RC extractor using PaddleOCR and an ordered token-pool detector that locks high-confidence fields first; ships with a Streamlit UI for live inspection and a batch CSV mode for bulk processing. [GitHub ↗](#)
- **Attention Monitor** • Delivered an offline attention & engagement monitor using MediaPipe FaceMesh and OpenCV solvePnP head-pose estimation; classifies attentive / looking-away / looking-up / slacking states with temporal smoothing, CPU-only. [GitHub ↗](#)

Data Scientist Intern

Jan 2025 – Oct 2025

Analytics Career Connect

Remote

- Performed RFM segmentation on 4,300+ customers, surfacing key loyalty cohorts and ~₹8.3M in revenue clusters; informed marketing focus and customer-targeting strategy. [GitHub ↗](#)
- Designed Power BI dashboards visualising purchase frequency, customer behaviour and revenue trends; cross-validated dashboard accuracy with SQL and Python checks.
- Owned end-to-end data preparation — cleaning, deduplication, missing-value handling and feature engineering — to ready datasets for downstream ML models.
- Applied statistical analysis, correlation checks and trend forecasting to surface operational inefficiencies and revenue opportunities; presented insights to cross-functional teams.

EDUCATION

MCA — Artificial Intelligence & Machine Learning

Aug 2023 – Aug 2025

Chandigarh University

India

BCA — Bachelor of Computer Applications

Jul 2020 – Jul 2023

VGC, Himachal Pradesh University

Mandi, India

CERTIFICATIONS

- Python 3 Programming — Coursera
- Web Development Bootcamp — Udemy
- Java Programming — Udemy